

**MATH 1342 – Elementary Statistics – Week 5: Linear Regression**

**Sep 22 – 24** | 2.5 Two Quantitative Variables: Linear Regression • Catch-Up Day (Thu Sep 24) | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

| Course-level Objectives                                                                              | Module-level Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Learning Materials                                                                      | #             | Activities                                                                                         | #                  | Assessments                                                                         | #    |
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| 5. Analyze relationships between two quantitative variables using correlation and linear regression. | 1. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CO5)<br>2. Use a regression equation to predict a response value. (CO5)<br>3. Compute and interpret a residual, and locate positive and negative residuals on a scatterplot. (CO5)<br>4. Use technology to find the least-squares regression line and explain what “least squares” means. (CO5)<br>5. Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CO5) | e-Text: 2.5 Two Quantitative Variables: Linear Regression                               | All           | Participation: Two Variable Relationships – Tue: correlation review, Your Turn: spotting residuals | MO3                | Online Homework: 2.5 Two Quantitative Variables:                                    | All  |
|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) | MO1, MO2, MO3 | Participation: Regression Line – Thu: baseball game length and used-car price problems             | MO1, MO2, MO3, MO5 | Linear Regression (due Sun Sep 27, 11:59 PM)                                        |      |
|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) | MO4, MO5      | Example 2 (commute distance and time): find the regression line with technology and predict        | MO2, MO4           | Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice) | None |
|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Slides: 2.5 Two Quantitative Variables: Linear Regression                               | All           |                                                                                                    |                    |                                                                                     |      |
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